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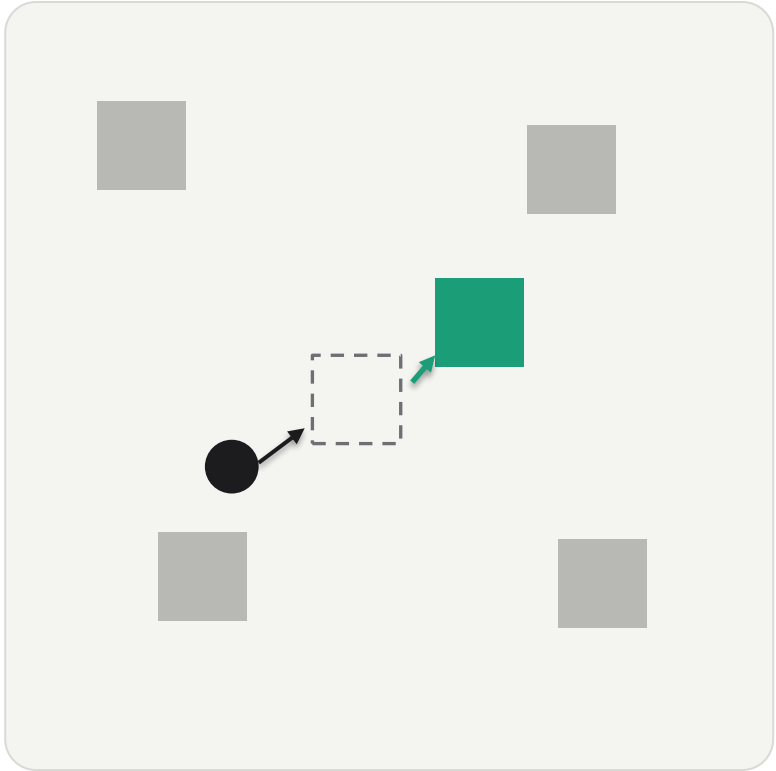
Modeling What Changes

Sparse, Residual World Models for Object-Centric Manipulation

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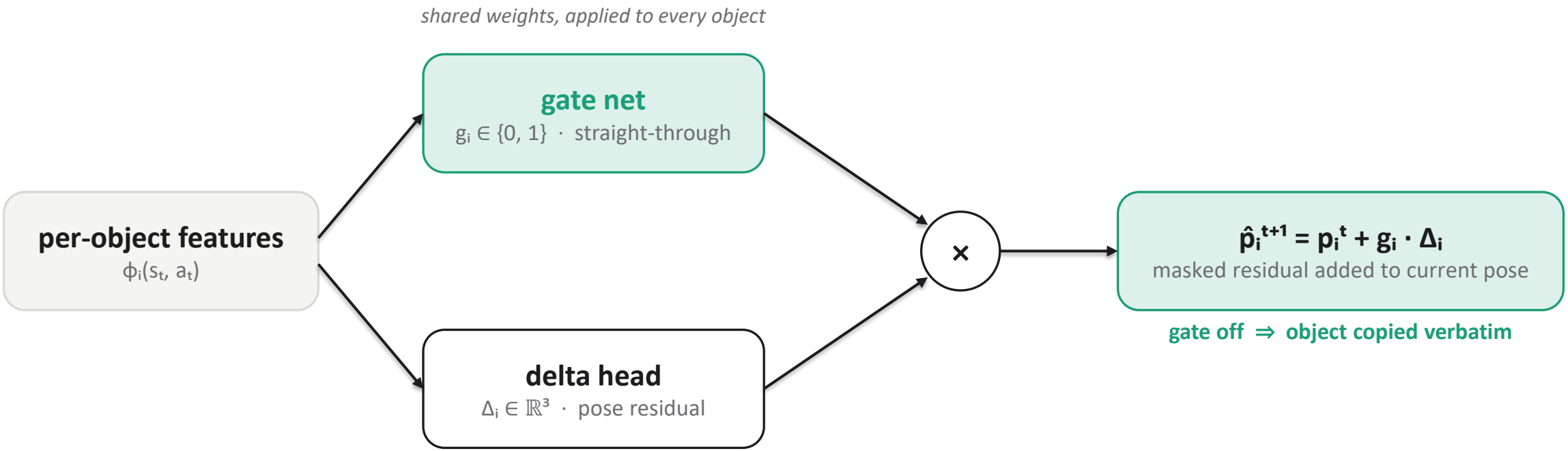
Manisha Sushant Gote ZuiGO Private Limited



THE IDEA

One push moves one object. So gate what changes, copy the rest.

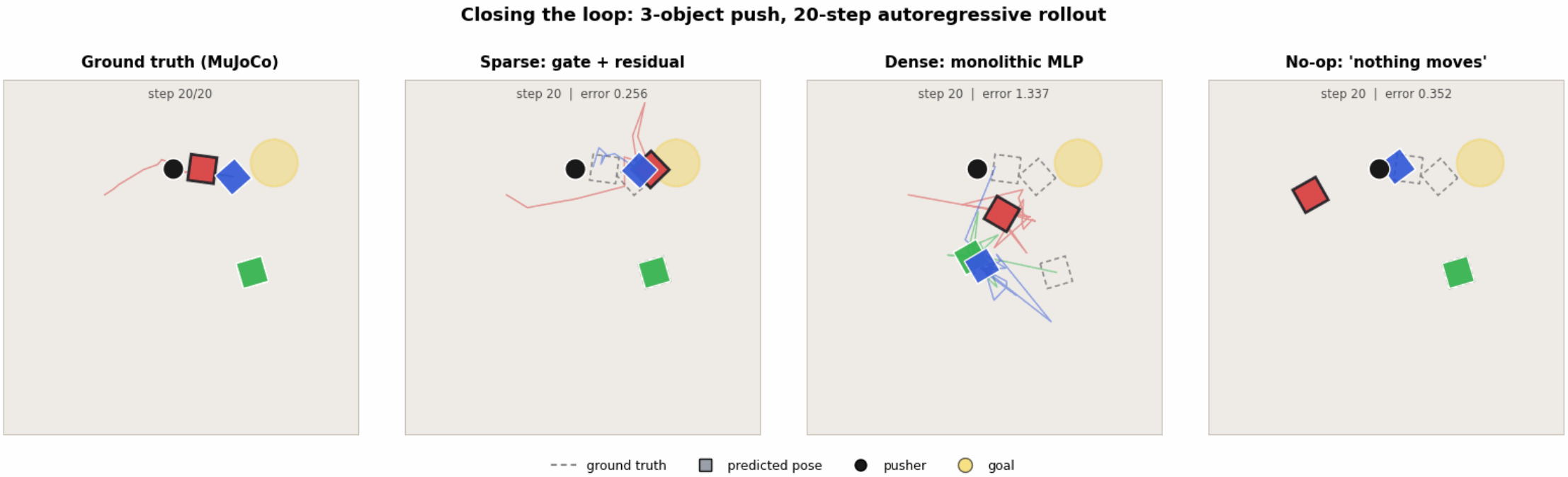
Dense world models regress *every* pose, every step, and so inject error into objects that never moved.



- **Hard gate at inference:** un-gated objects get exactly zero update
- **Shared per-object heads, nothing sized to N:** train on 3 objects, run on 8 with no retraining

SEE IT

Dense moves everything. Sparse moves only what was pushed.

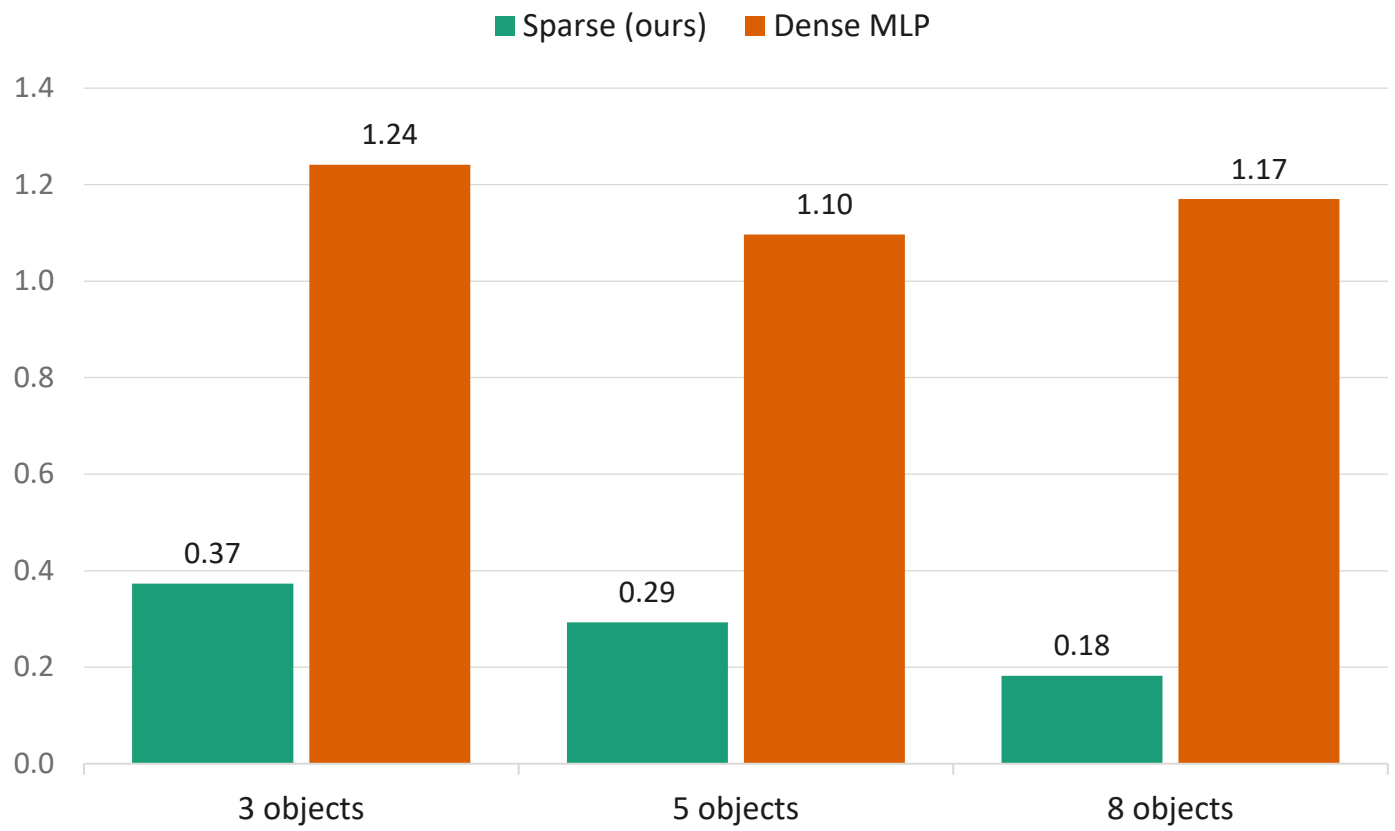


Held-out 3-object push, 20-step closed-loop rollout, half speed. Dashed outlines are the ground truth, so the gap is the error. Final error: sparse 0.26, dense 1.34, no-op 0.35.

Window picked by a fixed rule (median sparse error among active pushes), not by hand. Averaged over all rollouts, no-op is still lower overall (Table III).

RESULTS

Far less drift than dense, at a tenth of the parameters



Per-object pose L2 after a 20-step closed-loop rollout on held-out trajectories, lower is better (Table III, mean of 3 seeds). No-op, which predicts no motion, is lower still: 0.277 / 0.180 / 0.108.

2.5 – 4.6×

lower one-step error than dense, with 8.6–11.1× fewer parameters (Table I)

0.211 → 0.001

error on objects that did not move, dense → sparse (3 objects)

99.4%

of change-detection F1 kept on an unseen object count, no retraining

READING IT HONESTLY

What the headline is, and what it is not

IT IS

- **A change detector** (F1 0.80–0.87) that stops error injection into the static majority
- **Clearly better than dense** one-step, in rollout, and in planning (0.23 vs 0.00 success)

IT IS NOT

- **An accurate predictor of the movers:** an oracle gate barely helps; they sit at no-op quality
- **Better than predicting no motion, overall:** it only matches it. Planning is not clearly beyond random either

Lesson: always report the no-op baseline, and split error into changed vs. unchanged objects.

TAKEAWAY

Model what changes. The delta head is the lever.

- **A per-object gate plus a masked residual** is a one-line bias that stops a world model corrupting the static scene
- **Detection works;** regressing contact deltas does not yet. That is the next thing to fix
- **Everything is open:** code, data generators, checkpoints, and a live in-browser demo

Code + demo

github.com/ParamThakkar123/sparse_world_models

Come find us at the poster. Thank you!